

Nucleophilic Additions to Carbonyl Compounds

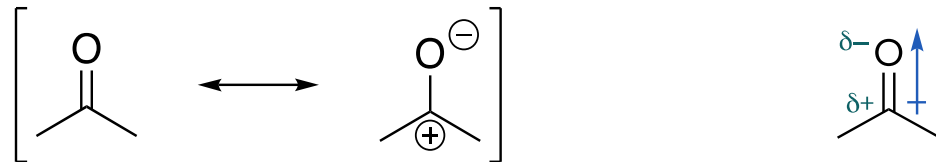
Organic Chemistry II

CHEM 2312

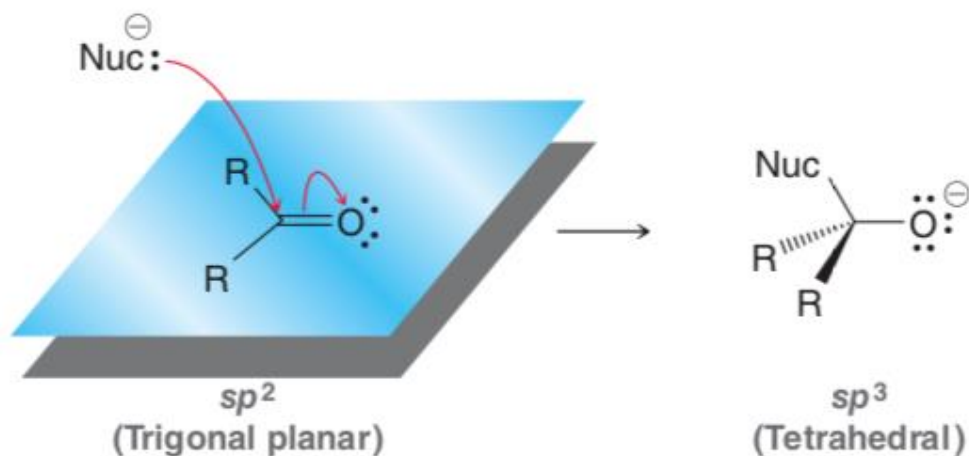
Dr. Michael Evans

1. Describe the general mechanism of nucleophilic addition to a carbonyl compound using elementary steps.
2. Predict the products of nucleophilic additions of organometallic compounds to ketohydes; describe the mechanisms of these reactions using elementary steps.
3. Apply the Wittig reaction in synthesis and retrosynthesis.
4. Predict the products of nucleophilic additions of oxygen nucleophiles (H_2O , ROH) to ketohydes; describe the mechanisms of these reactions using elementary steps.
5. Apply acetals as protecting groups in multi-step synthesis.
6. Predict the products of nucleophilic additions of nitrogen nucleophiles (H_2NR , HNR_2 , and related nitrogen nucleophiles) to ketohydes; describe the mechanisms of these reactions using elementary steps.
7. Predict the products of acidic hydrolysis of acetals, imines, enamines, and related condensation products of ketohydes with heteroatomic nucleophiles.

- The carbonyl carbon is electrophilic due to resonance and inductive effects

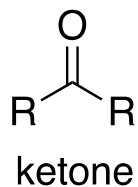
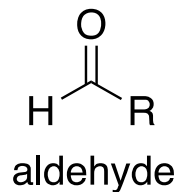


- Nucleophilic addition to the carbonyl carbon (Ad_N) is *the most important elementary step* in which carbonyl compounds engage

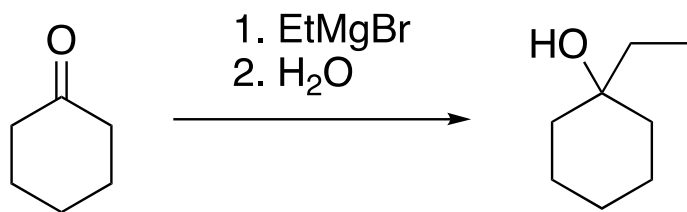


- Addition may establish a new stereocenter and generate a mixture of stereoisomers
- Beta-elimination (E_β) is the reverse of nucleophilic addition

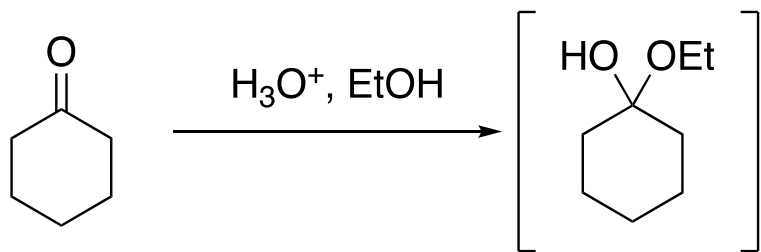
- Aldehydes are generally more reactive than ketones toward nucleophilic addition
 - Steric effect
 - Inductive effect



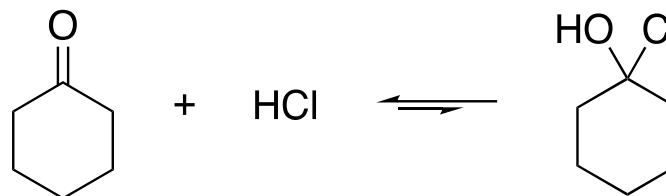
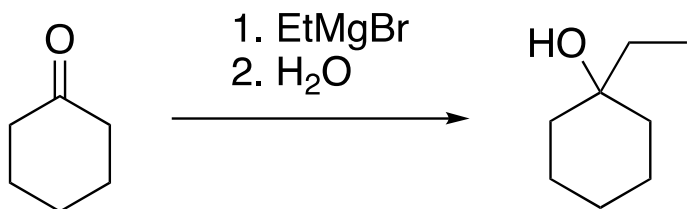
- Nucleophilic addition reactions may involve basic or acidic conditions
- Under basic conditions (organometallics), Ad_N is followed by proton transfer from water on aqueous workup



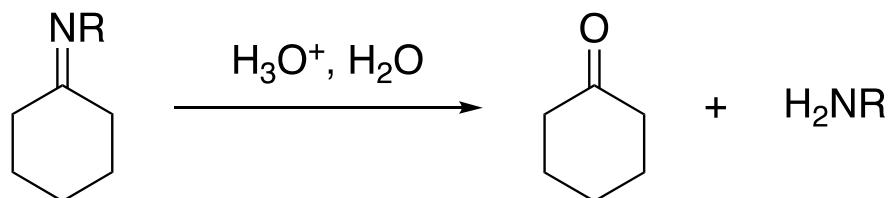
- Under acidic conditions (neutral nucleophile + acid), proton transfer to the carbonyl carbon is followed by Ad_N
- Note the protonated carbonyl intermediate!



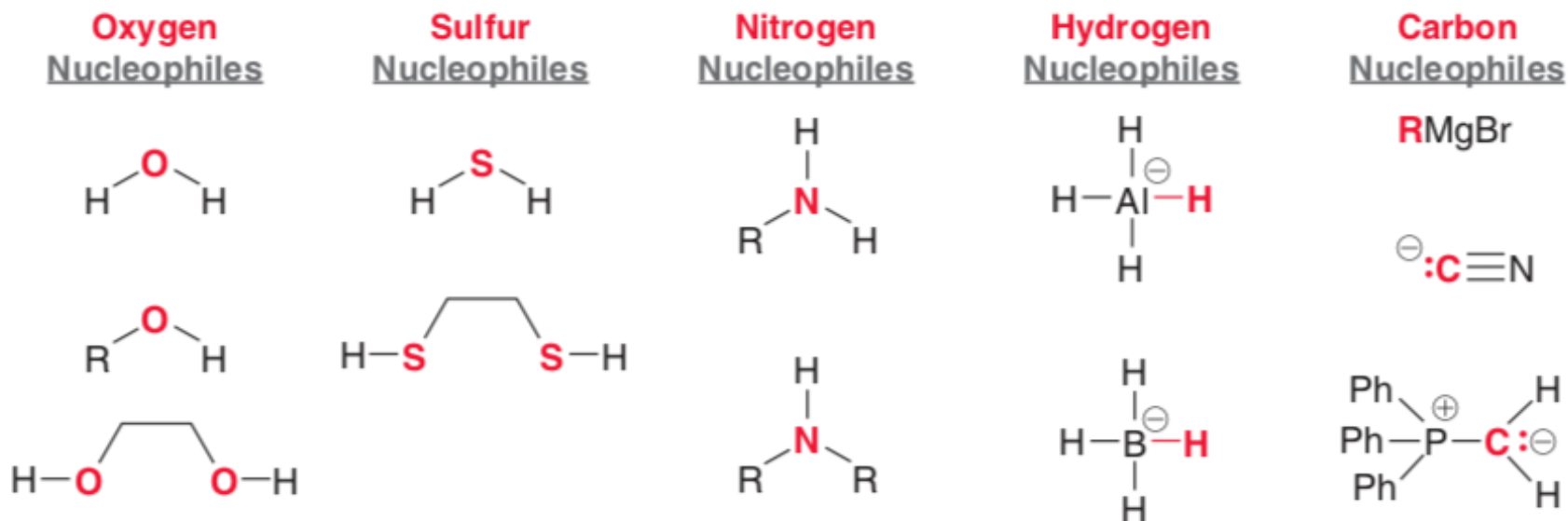
- Nucleophilic addition reactions may be reversible or irreversible, depending on the relative Lewis basicity and leaving group ability of the nucleophile



- Some (not all!) nucleophilic addition reactions can be reversed via treatment of the product with acidic water

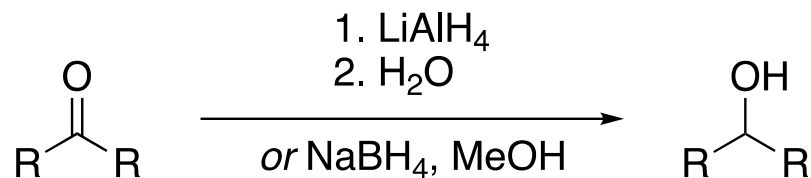


- Nucleophilic addition reactions may establish carbon–heteroatom, carbon–hydrogen, or carbon–carbon bonds

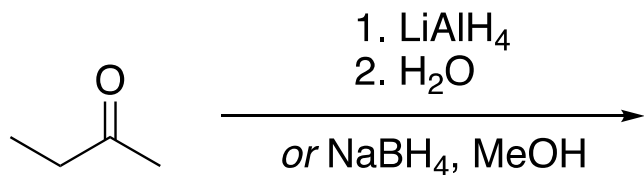


19.9 Hydrogen Nucleophiles (Reduction Revisited!)

- Complex metal hydrides add to ketones and aldehydes irreversibly to yield alcohols (section **12.4**)



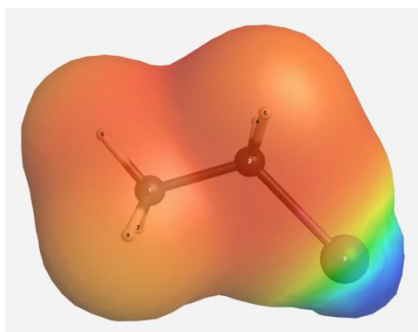
- Consider whether reduction of a ketone or aldehyde generates a new stereocenter



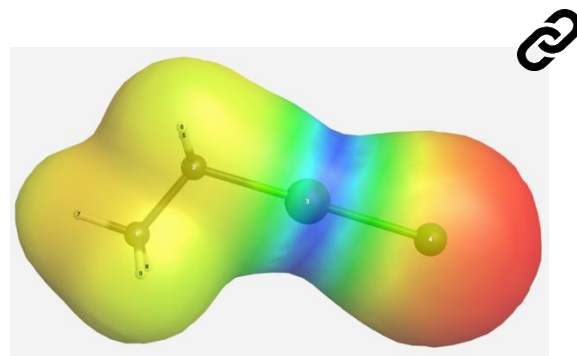
19.10 Introducing Organometallic Compounds

- **Organometallic compounds** contain at least one metal–carbon bond
- Because metals are electropositive with respect to carbon, carbon is a **nucleophile** in most of these species

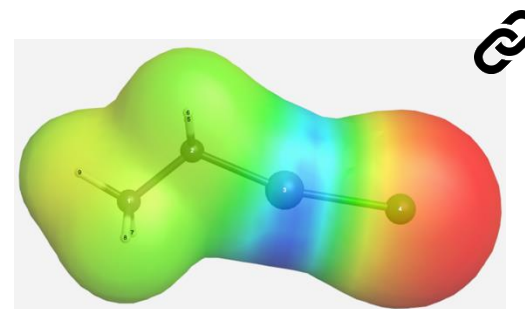
- The nucleophilicity of carbon depends on the ionic character of the bond
- The more electropositive the metal, the greater the nucleophilicity of the carbon



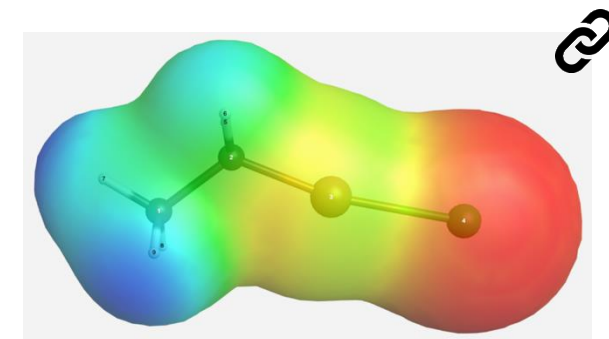
$\text{C}_2\text{H}_5\text{Li}$



$(\text{C}_2\text{H}_5)\text{MgCl}$



$(\text{C}_2\text{H}_5)\text{ZnCl}$



$(\text{C}_2\text{H}_5)\text{CuCl}^-$

C–Li C–Mg

C–Ti C–Ni

C–Cu C–Zn C–Hg

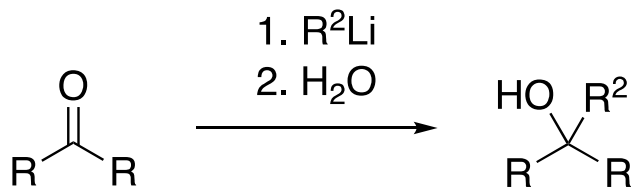
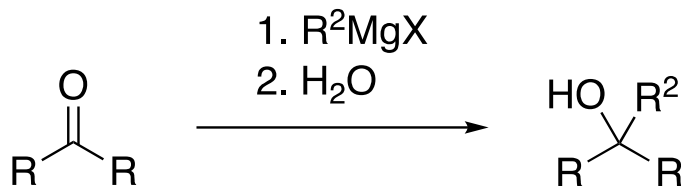
very strong nucleophiles

strong nucleophiles, ligand dependent

more tolerant of functional groups

19.10 Carbon Nucleophiles: Grignard Reagents and Organolithiums

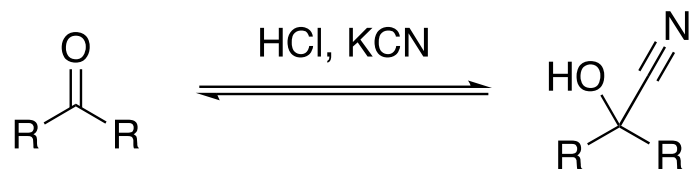
- A **Grignard reagent** is an organomagnesium halide (RMgX) formed via the reaction of an organohalide with magnesium metal (section 12.6)
- An **organolithium reagent** (RLi) can be prepared (1) from an organohalide and lithium metal, (2) via deprotonation with a strong base such as *n*-butyllithium or *tert*-butyllithium, or (3) via lithium-halogen exchange
- RLi and RMgX add to aldehydes and ketones irreversibly to yield alcohols after aqueous workup



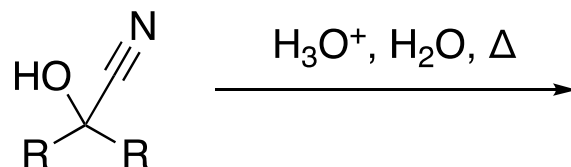
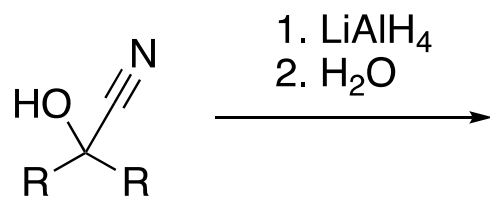
- Watch out for stereochemistry!

19.10 Cyanide Anion as a Nucleophile

- Treatment of a ketone or aldehyde with acid and an alkali metal cyanide (NaCN, KCN) results in reversible nucleophilic addition of hydrogen cyanide (HCN)

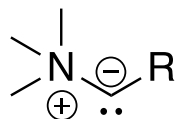


- The α -hydroxynitrile products can be further manipulated; for example using reduction or hydrolysis

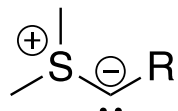


19.10 Ylides and the Wittig Reaction

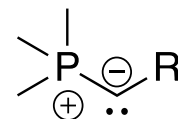
- **Ylides** contain a negatively charged atom (typically carbon) adjacent to a positively charged heteroatom
- Paradoxically, ylides are both nucleophilic and electrophilic



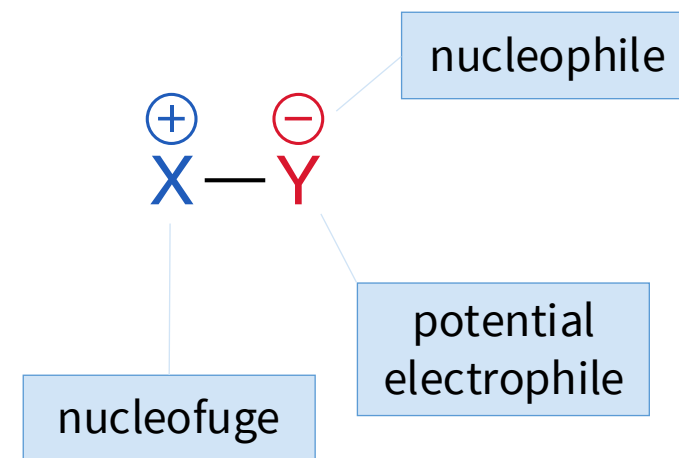
ammonium ylide



sulfonium ylide



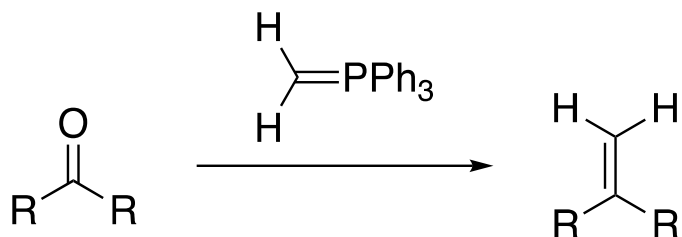
phosphonium ylide



- Alternative resonance forms suggest that ylides can participate in addition reactions

19.10 The Wittig Reaction

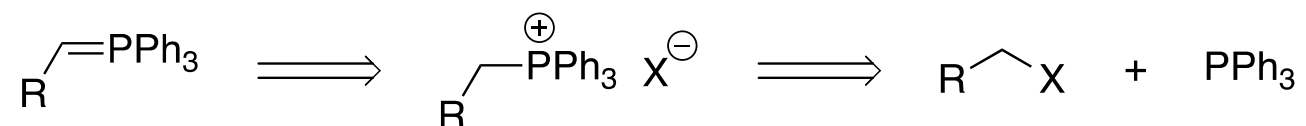
- Treatment of a ketone or aldehyde with a phosphonium ylide yields an alkene and triphenylphosphine oxide; this transformation is called the **Wittig reaction**



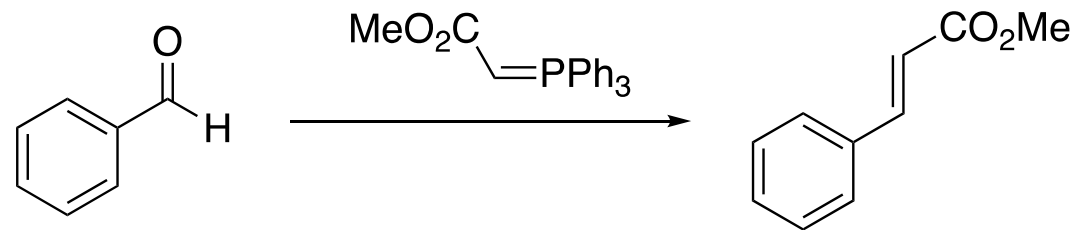
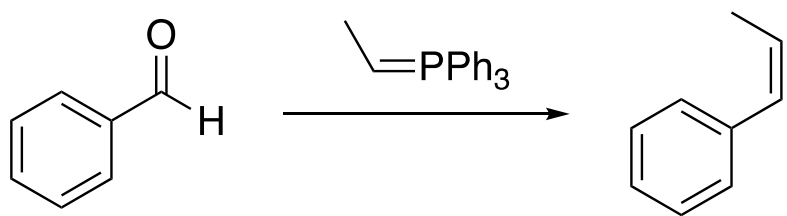
- The mechanism is most likely pericyclic: (2 + 2) cycloaddition is followed by retro-(2 + 2) elimination

19.10 The Wittig Reaction in Synthesis

- **Wittig reagents** (phosphonium ylides) are prepared from triphenylphosphine (PPh_3) using an $\text{S}_{\text{N}}2$ reaction followed by deprotonation with a very strong base ($n\text{-BuLi}$, PhLi , NaH , NaNH_2 , etc.)

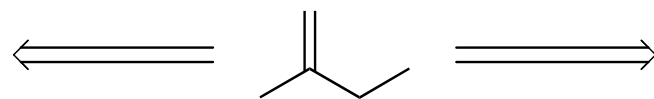


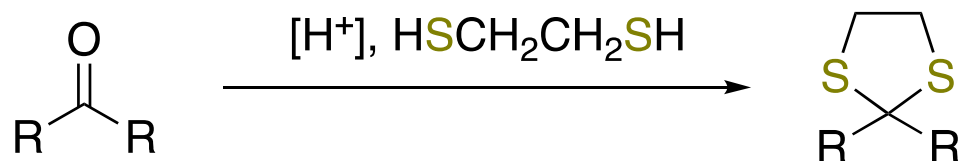
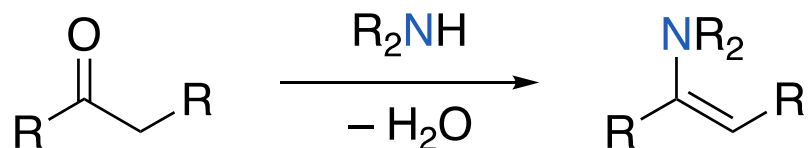
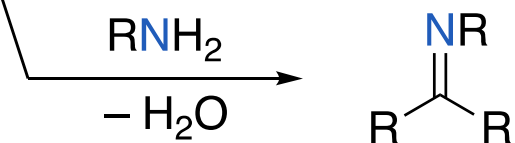
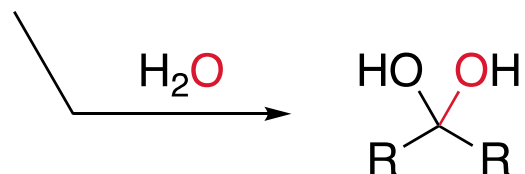
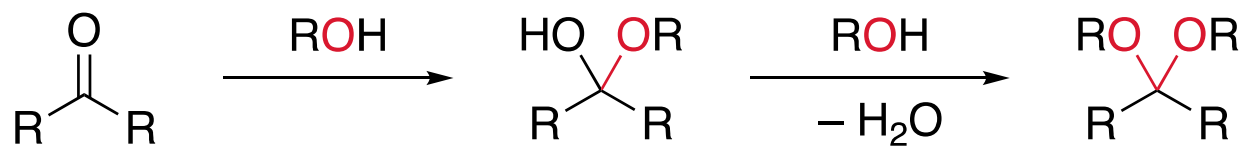
- Great synthetic utility; alkenes from carbonyl compounds (“reverse ozonolysis”!)
 - *Z*-selective when unstabilized Wittig reagents are used
 - *E*-selective when stabilized reagents are used



19.10 The Wittig Reaction in Retrosynthesis

- Most target alkenes can be associated with two distinct Wittig reactions
- The ideal choice is based on the ease of preparation of the Wittig reagent





Next, we focus on additions of heteroatomic nucleophiles (O, N, and S) to ketones and aldehydes. Nucleophilic addition (Ad_N) is the key elementary step in all these reactions. Some are accompanied by beta-elimination (E_β), which results in net substitution of the carbonyl oxygen. All these additions are **reversible** except in very specific circumstances.

vital strategy for organic synthesis

carbohydrates

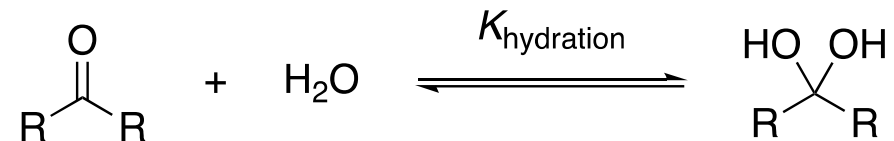
Maillard reactions

vitamin B₆

biosynthesis of aromatic heterocycles

19.5 Oxygen Nucleophiles: Hydration of Ketones and Aldehydes

- Dissolving a ketone or aldehyde in water results in **hydration** via a nucleophilic addition mechanism



- Carbonyl compound is generally favored at equilibrium, except for:
 - Very electrophilic ketones or aldehydes
 - Sterically unhindered aldehydes

Aldehydes	$K_{\text{hydration}}$	Ketones	$K_{\text{hydration}}$
HCHO	2.2×10^3	$(\text{CH}_3)_2\text{CO}$	1.4×10^{-3}
CH_3CHO	1.0	CH_3COPh	6.6×10^{-6}
$(\text{CH}_3)_2\text{CHCHO}$	0.75	Ph_2CO	1.2×10^{-7}
PhCHO	2.8×10^{-3}	$(\text{ClCH}_2)_2\text{CO}$	10
ClCH_2CHO	37	$(\text{CF}_3)_2\text{CO}$	∞
Cl_3CCHO	2.8×10^4		

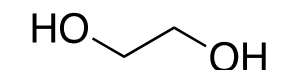
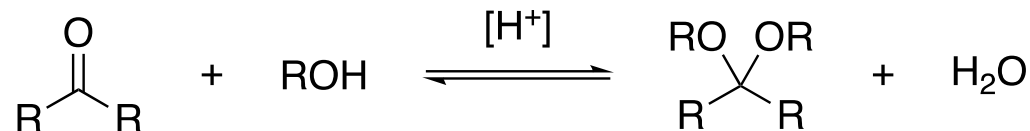
19.5 Hydration of Ketones and Aldehydes



- When catalyzed by acid, the ketohyde is first protonated (pt), nucleophilic addition follows (Ad_N), and a proton is transferred back to the conjugate base of the acid catalyst (pt)
 - *Under acidic conditions, reasonable mechanisms cannot include bases stronger than the conjugate base of the acid*
-
- When catalyzed by base, hydroxide adds to the ketohyde (pt) and a proton is transferred to the resulting alkoxide intermediate from water (pt)
 - *Under basic conditions, reasonable mechanisms cannot include acids stronger than the conjugate acid of the base*

19.5 Oxygen Nucleophiles: Acetal Formation

- Now, consider replacing water with an alcohol. Reaction with a ketone or aldehyde in the presence of an acid catalyst yields an **acetal** via a nucleophilic addition mechanism



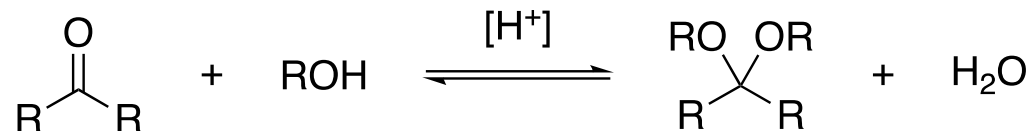
ethylene glycol

- Methanol, ethanol, and ethylene glycol ($\text{HOCH}_2\text{CH}_2\text{OH}$) are the most frequently used alcohols
- In the first stage of the mechanism, acid-catalyzed nucleophilic addition yields a **hemiacetal** intermediate (1. pt; 2. Ad_N; 3. pt)

- Hemiacetals appear frequently in biochemistry...

19.5 Acetal Formation

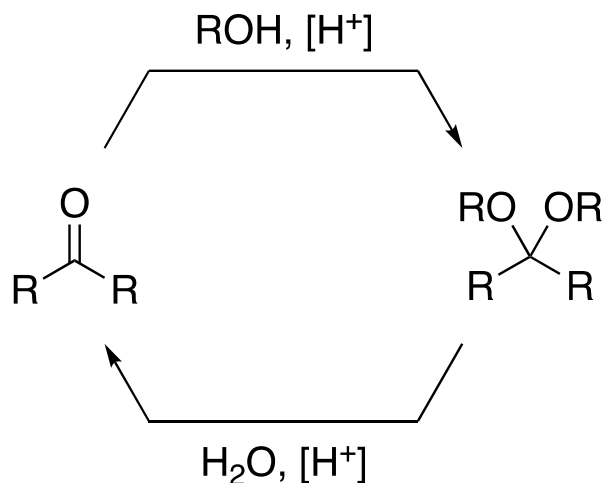
- Now, consider replacing water with an alcohol. Reaction with a ketone or aldehyde in the presence of an acid catalyst yields an **acetal** via a nucleophilic addition mechanism



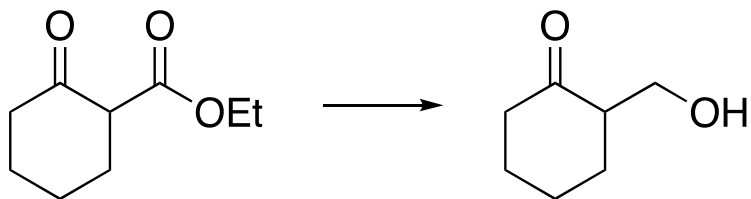
- In the second stage of the mechanism, acid-catalyzed $\text{S}_{\text{N}}1$ reaction yields the final acetal product (4. pt; 5. E_{β} ; 6. Ad_{N} ; 7. pt)

19.5 Acetals as Protecting Groups in Synthesis

- Acetals can be converted back to ketohydes via acidic **hydrolysis**
- Acetals are **indefinitely stable** under strongly basic/nucleophilic conditions!
- Thus, acetals can be used as **protecting groups** for ketones or aldehydes under strongly basic conditions

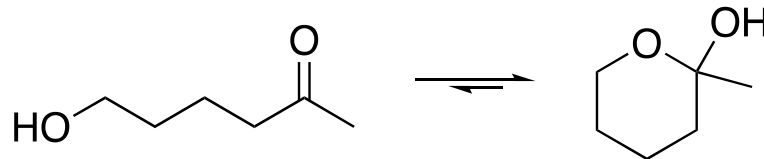


- For example, let's plan the following synthesis

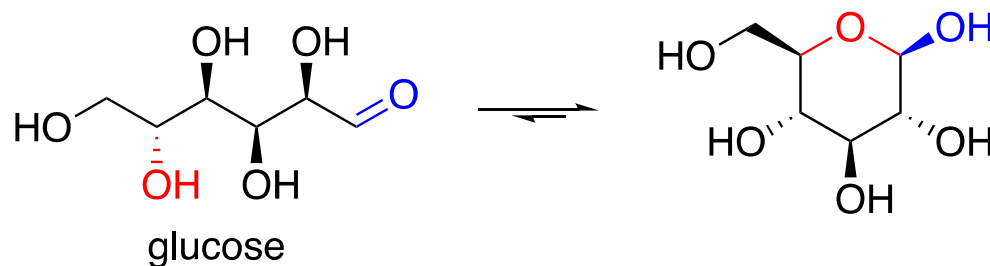


19.5 Hydroxy Ketohydes Can Form Stable Cyclic Hemiacetals

- Hydroxyketones or -aldehydes may form a stable cyclic hemiacetal if the ring is five- or six-membered

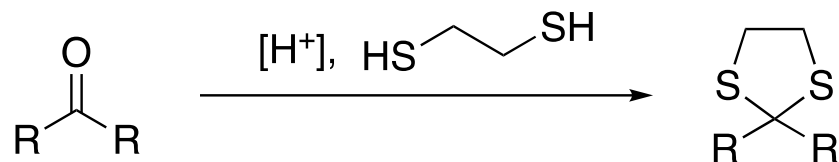


- Cyclic hemiacetals are the stable forms of monosaccharides (more later in CHEM 2312 LS!)

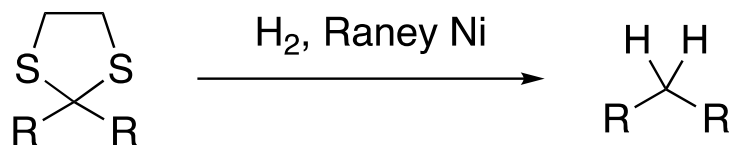


19.8 Sulfur Nucleophiles: Thioacetal Formation

- Sulfur nucleophiles react analogously to alcohols with ketones and aldehydes, forming **thioacetals**

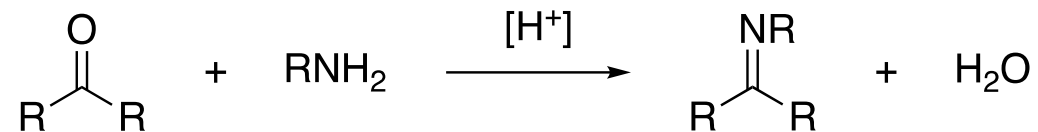


- Raney nickel can be used to reduce the thioacetal carbon to a methylene ($-\text{CH}_2-$); the two-step sequence is an alternative to the Clemmensen and Wolff-Kishner reactions



19.6 Nitrogen Nucleophiles: Imines from Primary Amines

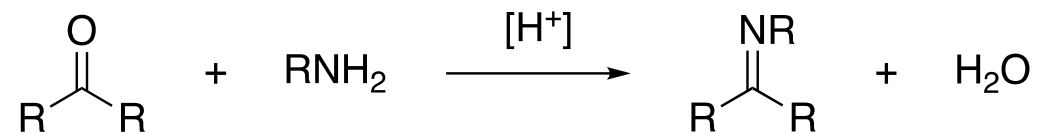
- Primary amines (RNH_2) condense with ketohydes to form **imines (Schiff bases)** with loss of water



- It is essential to use a weak acid, most often acetic acid (HOAc)
- In the first stage of the mechanism, acid-catalyzed nucleophilic addition yields a **carbinolamine** intermediate (1. Ad_N ; 2. pt; 3. pt). *Note the order of steps!*

19.6 Imines from Primary Amines

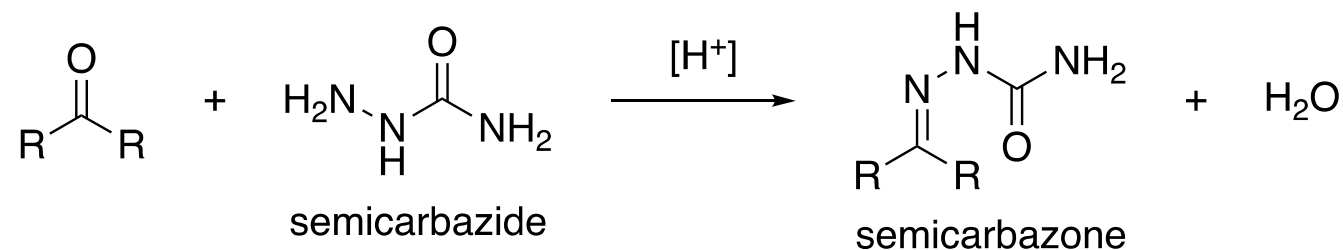
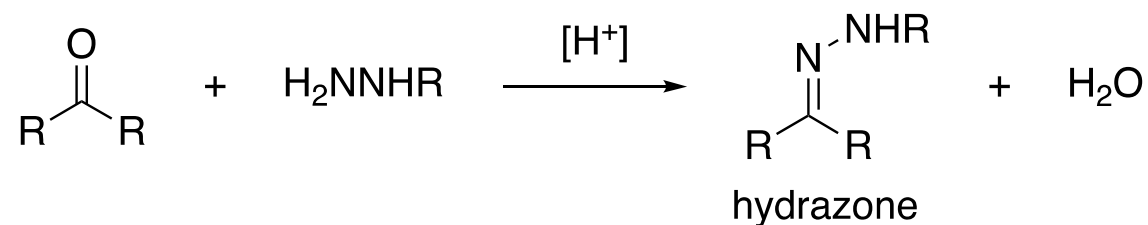
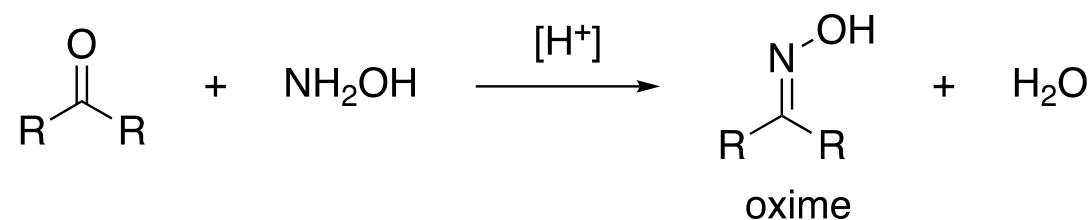
- Primary amines (RNH_2) condense with ketohydes to form **imines (Schiff bases)** with loss of water



- In the second stage of the mechanism, acid-catalyzed beta-elimination yields the imine and a molecule of water via an **iminium ion** intermediate (1. pt; 2. E_β ; 3. pt)

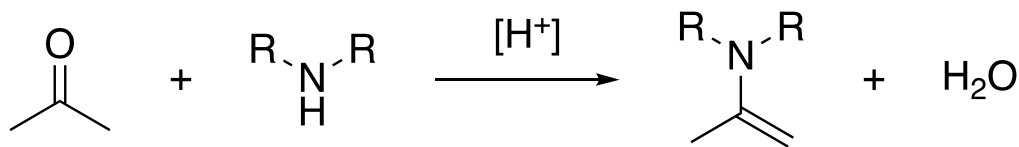
19.6 Forming Oximes, Hydrazones, and More

- Other nitrogen nucleophiles containing an -NH_2 group (NH_2OH , NH_2NHR , etc.) condense similarly with ketohydes to give C=N-containing products



19.6 Enamines from Secondary Amines | 1 of 2

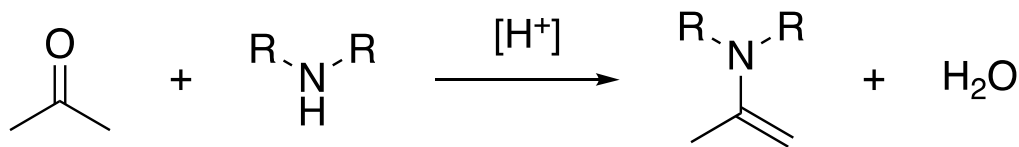
- Secondary amines (R_2NH) condense with ketohydes to form **enamines** with loss of water



- It is essential to use a weak acid, most often acetic acid (HOAc)
- In the first stage of the mechanism, acid-catalyzed nucleophilic addition yields a **carbinolamine** intermediate (1. Ad_N ; 2. pt; 3. pt). *Note the order of steps!*

19.6 Enamines from Secondary Amines | 2 of 2

- Secondary amines (R_2NH) condense with ketohydes to form **enamines** with loss of water

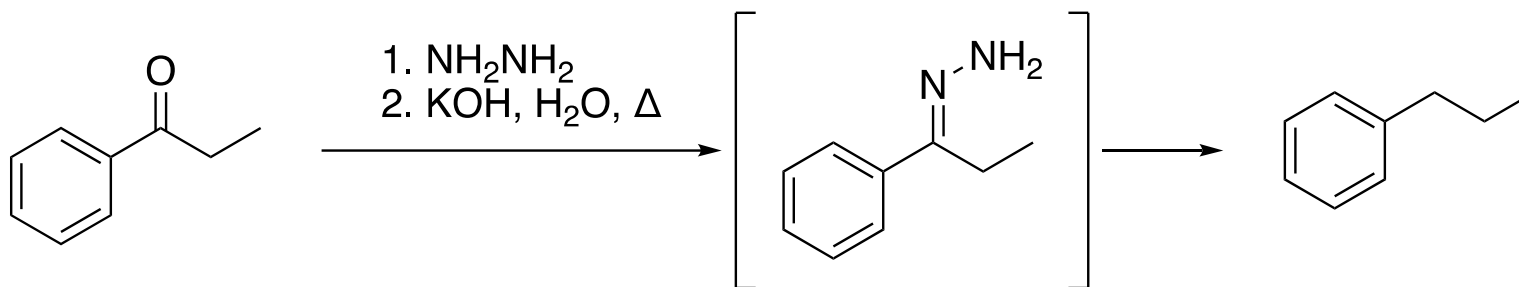


- In the second stage of the mechanism, acid-catalyzed beta-elimination yields the enamine and a molecule of water via an **iminium ion** intermediate (1. pt; 2. E_β ; 3. pt)

- We will only study enamine condensations of aldehydes and symmetrical ketones (no regiochemical issues)

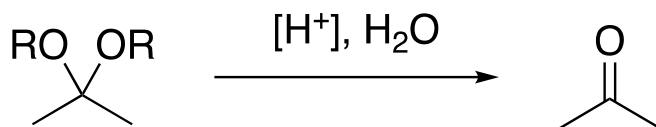
19.6 The Wolff-Kishner Reduction Revisited

- The Wolff-Kishner reduction involves hydrazone formation followed by irreversible elimination of dinitrogen

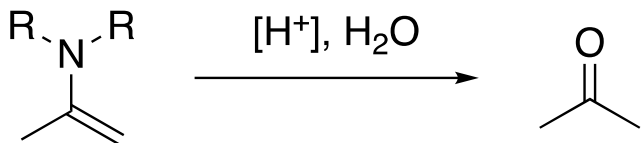
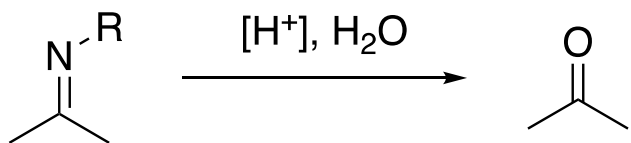


- Following formation of the hydrazone, three proton transfers (1. pt; 2. pt; 3. pt) set up the elimination of N_2 (4. E_β) and a final proton transfer follows (5. pt)

- All the above nucleophilic addition reactions can be reversed by treating the product with acidic water
- Bonds are cleaved through the action of water as a nucleophile, so this process is called **hydrolysis**
- Hydrolytic mechanisms are the **microscopic reverses** of the mechanisms we have already seen!
- For example, for acidic hydrolysis of an acetal,



- All the above nucleophilic addition reactions can be reversed by treating the product with acidic water
- Bonds are cleaved through the action of water as a nucleophile, so this process is called **hydrolysis**
- Hydrolytic mechanisms are the **microscopic reverses** of the mechanisms we have already seen!
- Imines, enamines, and other condensation products also undergo hydrolysis when dissolved in acidic water



19.7 Drawing Products of Hydrolysis Reactions

- Identify the bond broken via hydrolysis (C-X; X = N, O)
- Identify the future carbonyl carbon, which is at the oxidation level of a ketone or aldehyde
- Draw the products expected for each of the following reactions:

